

AVERAGE FIRST-PASSAGE TIMES FOR CHARACTER SUMS

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ABSTRACT. Let $\varepsilon > 0$ and, for an odd prime p , set

$$S_\ell(p) := \sum_{n \leq \ell} \left(\frac{n}{p} \right).$$

We solve Problem #981 on the Erdős Problems website concerning the *first-passage time*

$$f_\varepsilon(p) := \min\{\ell \geq 1 : S_\ell(p) < \varepsilon \ell\}.$$

We prove that there exists a constant $c_\varepsilon > 0$ such that

$$\sum_{p \leq x} f_\varepsilon(p) \sim c_\varepsilon \frac{x}{\log x}$$

as $x \rightarrow \infty$.

1. INTRODUCTION

A central question in number theory is to understand the behavior of the quadratic character sum

$$S_\ell(p) := \sum_{n \leq \ell} \left(\frac{n}{p} \right),$$

where $\left(\frac{\cdot}{p} \right)$ denotes the Legendre symbol modulo an odd prime p . Such sums encode fundamental information on the distribution of quadratic residues and non-residues modulo p . In particular, bounds for $S_\ell(p)$ lead to results on the size of the least quadratic non-residue modulo p ; for a more detailed discussion and further background, see [6, Section 1].

Let $\chi_p(n) := \left(\frac{n}{p} \right)$. A basic heuristic models the values $\chi_p(n)$ as independent unbiased ± 1 signs, in which case $S_\ell(p)$ is typically of size $\asymp \sqrt{\ell}$ and, for any fixed $\varepsilon > 0$, the inequality $S_\ell(p) < \varepsilon \ell$ should hold for all sufficiently large ℓ . This motivates quantitative questions about the time at which the graph of $S_\ell(p)$ falls below the linear barrier $\varepsilon \ell$.

In [2, Eq. (80)], Erdős posed the following conjecture.

Conjecture 1.1 (Erdős's eventual-time problem). *Let $\varepsilon > 0$, and for each odd prime p let $F_\varepsilon(p)$ be the smallest integer such that*

$$S_\ell(p) < \varepsilon \ell \quad \text{for every } \ell \geq F_\varepsilon(p). \tag{1.1}$$

Then there exists a constant $C_\varepsilon > 0$ such that

$$\sum_{p \leq x} F_\varepsilon(p) \sim C_\varepsilon \frac{x}{\log x}.$$

This conjecture was proved by Elliott [3].

A related variant appears as Problem #981 on Bloom's Erdős Problems website [1].

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Conjecture 1.2 (Erdős's first-passage time problem). *Let $\varepsilon > 0$, and for each odd prime p let $f_\varepsilon(p)$ be the smallest integer k such that $S_k(p) < \varepsilon k$. Then there exists a constant $c_\varepsilon > 0$ such that*

$$\sum_{p \leq x} f_\varepsilon(p) \sim c_\varepsilon \frac{x}{\log x}.$$

Clearly one has $f_\varepsilon(p) \leq F_\varepsilon(p)$ for each p , since (1.1) forces $S_{F_\varepsilon(p)}(p) < \varepsilon F_\varepsilon(p)$. However, an asymptotic formula for $\sum_{p \leq x} F_\varepsilon(p)$ does not by itself imply the corresponding asymptotic for $\sum_{p \leq x} f_\varepsilon(p)$. To the best of our knowledge, Conjecture 1.2 was previously open. In this paper we solve it.

Theorem 1.3. *For every $\varepsilon > 0$ there exists a constant $c_\varepsilon \in (0, \infty)$ such that, as $x \rightarrow \infty$,*

$$\sum_{p \leq x} f_\varepsilon(p) \sim c_\varepsilon \frac{x}{\log x}.$$

2. PRELIMINARIES

2.1. Notation and conventions. Fix $\varepsilon > 0$. The case $\varepsilon > 1$ is trivial since $S_1(p) = 1 < \varepsilon$ for every odd prime p , hence $f_\varepsilon(p) = 1$. Thus, throughout the paper we assume that $0 < \varepsilon \leq 1$, with the convention $S_0(p) := 0$.

We use Vinogradov's asymptotic notation. For functions $f = f(n)$ and $g = g(n)$, we write $f = O(g)$, $f \ll g$, or $g \gg f$ to mean that there exists a constant $C > 0$ such that $|f(n)| \leq Cg(n)$ for all sufficiently large n . We write $f \asymp g$ to mean that $f \ll g$ and $g \ll f$, and we write $f = o(g)$ to mean that $f(n)/g(n) \rightarrow 0$ as $n \rightarrow \infty$. If the implicit constant is allowed to depend on one or more parameters $z_1, \dots, z_r$, we indicate this by writing $f \ll_{z_1, \dots, z_r} g$, $g \gg_{z_1, \dots, z_r} f$ or $f = O_{z_1, \dots, z_r}(g)$. Finally, we write $f \sim g$ to mean that $f(n)/g(n) \rightarrow 1$ as $n \rightarrow \infty$.

Throughout the paper, all primes are assumed to be odd unless stated otherwise. Accordingly, $\pi(x)$ denotes the number of odd primes $\leq x$, and $\sum_{p \leq x}$ means a sum over odd primes. We denote by $\mathcal{F}(x)$ the set of fundamental discriminants d such that $|d| \leq x$. We write $mn = \square$ to indicate that mn is a perfect square.

2.2. Two forms of the quadratic large sieve. Next we present two forms of the quadratic large sieve. The first is a large sieve inequality for prime discriminants, which is a special case of [9, Lemma 9]; see also [6, Lemma 4.8].

Lemma 2.1 ([9]). *Let $\kappa > 0$ be a fixed small number. Let $x \geq 2$ and $2 \leq N \leq x^{1/2-\kappa}$. Then for arbitrary complex numbers $a_1, \dots, a_N$ we have*

$$\sum_{p \leq x} \left| \sum_{n \leq N} a_n \left(\frac{n}{p} \right) \right|^2 \ll_\kappa \frac{x}{\log x} \sum_{\substack{m, n \leq N \\ mn = \square}} |a_m a_n|.$$

The second form is due to Heath–Brown [5, Corollary 2]; see also [8, Lemma 4.1].

Lemma 2.2 ([5]). *Let $x, N \geq 2$. Then for arbitrary complex numbers $a_1, \dots, a_N$ and for any $\rho > 0$ we have*

$$\sum_{d \in \mathcal{F}(x)} \left| \sum_{n \leq N} a_n \left(\frac{d}{n} \right) \right|^2 \ll_\rho (xN)^\rho (x+N) \sum_{\substack{m, n \leq N \\ mn = \square}} |a_m a_n|.$$

Here $\left(\frac{d}{\cdot} \right)$ denotes the Kronecker symbol modulo $|d|$.

2.3. Cubic coefficients and a square-condition bound. For an integer $\ell \geq 1$, define

$$c_u(\ell) := \#\{(a, b, c) \in \{1, \dots, \ell\}^3 : abc = u\} \quad (1 \leq u \leq \ell^3), \quad (2.1)$$

and set

$$C_3(\ell) := \sum_{\substack{u, v \leq \ell^3 \\ uv = \square}} c_u(\ell) c_v(\ell). \quad (2.2)$$

Lemma 2.3. *For every $\delta > 0$ one has*

$$C_3(m) \ll_{\delta} m^{3+\delta}.$$

Proof. We use the trivial bound $c_u(m) \leq d_3(u)$, where

$$d_k(u) = \#\{(a_1, \dots, a_k) \in \mathbb{N}^k : a_1 \cdots a_k = u\}.$$

To bound $d_3(u)$ in terms of the usual divisor function $d(u) = d_2(u)$. Write

$$u = \prod_p p^{s_p}.$$

Then

$$d_3(u) = \prod_{p|u} \#\{(e_1, e_2, e_3) \in \mathbb{N}^3 : e_1 + e_2 + e_3 = s_p\} = \prod_{p|u} \binom{s_p + 2}{2}.$$

Since $\binom{s+2}{2} = \frac{(s+1)(s+2)}{2} \leq (s+1)^2$ for every $s \geq 0$, we obtain

$$d_3(u) \leq \prod_{p|u} (s_p + 1)^2 = d(u)^2 \ll_{\theta} u^{2\theta}.$$

The last inequality follows from a classical result that $d(n) \ll_{\theta} n^{\theta}$ (see e.g. [4, Theorem 315]). Thus $c_u(m) \ll_{\theta} u^{2\theta}$.

The condition $uv = \square$ is equivalent to writing $u = ds^2$ and $v = dt^2$ with d squarefree. Therefore,

$$C_3(m) \leq \sum_{\substack{d \text{ squarefree} \\ d \leq m^3}} \left(\sum_{ds^2 \leq m^3} c_{ds^2}(m) \right)^2 \ll_{\theta} \sum_{d \leq m^3} \left(\sum_{s \leq m^{3/2}/\sqrt{d}} (ds^2)^{2\theta} \right)^2.$$

The inner sum satisfies

$$\sum_{s \leq m^{3/2}/\sqrt{d}} (ds^2)^{2\theta} = d^{2\theta} \sum_{s \leq m^{3/2}/\sqrt{d}} s^{4\theta} \leq d^{2\theta} \left(\frac{m^{3/2}}{\sqrt{d}} \right)^{1+4\theta} = m^{\frac{3}{2}(1+4\theta)} d^{-\frac{1}{2}}.$$

Hence

$$C_3(m) \ll_{\theta} \sum_{d \leq m^3} \left(m^{\frac{3}{2}(1+4\theta)} d^{-\frac{1}{2}} \right)^2 = m^{3+12\theta} \sum_{d \leq m^3} d^{-1} \ll_{\theta} m^{3+12\theta} \log m.$$

Now choose $\theta = \delta/24$ and use $\log m \ll_{\delta} m^{\delta/2}$ to conclude $C_3(m) \ll_{\delta} m^{3+\delta}$. $\square$

Next we give a uniform bound for $f_{\varepsilon}(p)$ via the Pólya–Vinogradov inequality.

Lemma 2.4. *There exists an absolute constant $C > 0$ such that for every odd prime p and every $\ell \geq 1$,*

$$|S_{\ell}(p)| < C\sqrt{p} \log p.$$

Consequently,

$$f_{\varepsilon}(p) \leq \frac{C}{\varepsilon} \sqrt{p} \log p$$

for all odd primes p .

Proof. The Pólya–Vinogradov inequality (see, for example [10, Theorem 9.18]), states that for any nonprincipal Dirichlet character χ modulo q ,

$$\max_{1 \leq t \leq q} \left| \sum_{n \leq t} \chi(n) \right| \ll \sqrt{q} \log q.$$

Applying this to $\chi = \chi_p$ gives $|S_\ell(p)| \ll \sqrt{p} \log p$ for $1 \leq \ell \leq p$. For $\ell > p$, the character χ_p is periodic modulo p and $\sum_{n=1}^p \chi_p(n) = 0$, so $S_\ell(p) = S_{\ell \bmod p}(p)$ and the same bound holds. Hence, there exists an absolute constant $C > 0$ such that for every odd prime p and every $\ell \geq 1$, $|S_\ell(p)| < C\sqrt{p} \log p$. For any $\ell \geq (C/\varepsilon)\sqrt{p} \log p$, we have $S_\ell(p) \leq |S_\ell(p)| < \varepsilon \ell$, so $f_\varepsilon(p) \leq \frac{C}{\varepsilon} \sqrt{p} \log p$. $\square$

2.4. A density parameter. Now, we need to estimate the density of primes such that $S_\ell(p) \geq \varepsilon \ell$. For this purpose, for $x \geq 2$ and each integer $\ell \geq 1$, we define

$$P_\ell(x) := \frac{1}{\pi(x)} \#\{p \leq x : S_\ell(p) \geq \varepsilon \ell\}. \quad (2.3)$$

We give the estimation of $P_\ell(x)$ through the sixth-moment of $S_\ell(p)$.

Lemma 2.5. *Fix $\kappa \in (0, 1/6)$ and $\delta \in (0, 1)$. When $2 \leq \ell \leq x^{1/6-\kappa}$, we have the uniform bound*

$$P_\ell(x) \ll_{\varepsilon, \kappa, \delta} \ell^{-3+\delta}. \quad (2.4)$$

When $\ell > x^{1/6-\kappa}$, we have the uniform bound

$$P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^{\rho-1} \ell^{3\rho-3+\delta} (x + \ell^3). \quad (2.5)$$

Proof. Using the following trivial bound, we transform the problem to the estimation of sixth-moment of $S_\ell(p)$.

$$P_\ell(x) \leq \frac{1}{\varepsilon^6 \ell^6 \pi(x)} \sum_{p \leq x} |S_\ell(p)|^6. \quad (2.6)$$

By (2.1), we have the identity

$$S_\ell(p)^3 = \sum_{a, b, c \leq \ell} \left(\frac{abc}{p} \right) = \sum_{u \leq \ell^3} c_u(\ell) \left(\frac{u}{p} \right).$$

Therefore

$$|S_\ell(p)|^6 = \left| \sum_{u \leq \ell^3} c_u(\ell) \left(\frac{u}{p} \right) \right|^2. \quad (2.7)$$

Case 1: $2 \leq \ell \leq x^{1/6-\kappa}$. Then $\ell^3 \leq x^{1/2-3\kappa}$. Applying Lemma 2.1 with $N = \ell^3 \geq 2$, we obtain

$$\sum_{p \leq x} |S_\ell(p)|^6 = \sum_{p \leq x} \left| \sum_{u \leq \ell^3} c_u(\ell) \left(\frac{u}{p} \right) \right|^2 \ll_{\kappa} \frac{x}{\log x} \sum_{\substack{u, v \leq \ell^3 \\ uv = \square}} c_u(\ell) c_v(\ell) = \frac{x}{\log x} C_3(\ell).$$

By Lemma 2.3, $C_3(\ell) \ll_{\delta} \ell^{3+\delta}$, hence

$$\sum_{p \leq x} |S_\ell(p)|^6 \ll_{\kappa, \delta} \frac{x}{\log x} \ell^{3+\delta}. \quad (2.8)$$

Since $\pi(x) \asymp x / \log x$, inserting (2.8) into (2.6) yields (2.4).

Case 2: $\ell > x^{1/6-\kappa}$. Let $\rho > 0$ be a small parameter. For a prime p , define $p^* := (-1)^{(p-1)/2}p$. Quadratic reciprocity implies $\left(\frac{u}{p}\right) = \left(\frac{p^*}{u}\right)$ for all integers u . Thus

$$\sum_{p \leq x} |S_\ell(p)|^6 = \sum_{p \leq x} \left| \sum_{u \leq \ell^3} c_u(\ell) \left(\frac{p^*}{u}\right) \right|^2 \leq \sum_{d \in \mathcal{F}(x)} \left| \sum_{u \leq \ell^3} c_u(\ell) \left(\frac{d}{u}\right) \right|^2.$$

Applying Lemma 2.2 with $N = \ell^3$, we obtain

$$\begin{aligned} \sum_{p \leq x} |S_\ell(p)|^6 &\ll_\rho (x\ell^3)^\rho (x + \ell^3) \sum_{\substack{u, v \leq \ell^3 \\ uv = \square}} c_u(\ell) c_v(\ell) = (x\ell^3)^\rho (x + \ell^3) C_3(\ell) \\ &\ll_{\rho, \delta} (x\ell^3)^\rho (x + \ell^3) \ell^{3+\delta}. \end{aligned} \quad (2.9)$$

Combining (2.9) with (2.6) and $\pi(x) \asymp x/\log x$ gives (2.5). $\square$

As a consequence, we obtain a uniform upper bound for $P_\ell(x)$.

Theorem 2.6. *Fix $\kappa \in (0, \frac{1}{6})$. Then for any positive real number $\beta < \frac{1}{3} - 2\kappa$, we have*

$$\sum_{\ell > x^{1/6-\kappa}} P_\ell(x) \ll_{\varepsilon, \kappa, \beta} x^{-\beta}. \quad (2.10)$$

Proof. Let $C > 0$ be the absolute constant from Lemma 2.4. For primes $p \leq x$ and integers $\ell \geq (C/\varepsilon)\sqrt{x} \log x$, Lemma 2.4 gives $|S_\ell(p)| < C\sqrt{x} \log x \leq \varepsilon\ell$, hence $P_\ell(x) = 0$. Thus

$$\sum_{\ell > x^{1/6-\kappa}} P_\ell(x) = \sum_{x^{1/6-\kappa} < \ell \leq (C/\varepsilon)\sqrt{x} \log x} P_\ell(x).$$

For any $\ell \geq x^{1/6-\kappa}$, (2.5) yields the bound

$$P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^{\rho-1} \ell^{3\rho-3+\delta} (x + \ell^3) \quad (\ell \geq 2). \quad (2.11)$$

Split the summation at $\ell = x^{1/3}$.

Range I: $x^{1/6-\kappa} < \ell \leq x^{1/3}$. Here $x + \ell^3 \leq 2x$, so (2.11) gives

$$P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^\rho \ell^{3\rho-3+\delta}.$$

Assuming $3\rho - 2 + \delta < 0$, we obtain

$$\sum_{x^{1/6-\kappa} < \ell \leq x^{1/3}} P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^\rho \sum_{\ell > x^{1/6-\kappa}} \ell^{3\rho-3+\delta} \ll_{\varepsilon, \rho, \delta} (\log x) x^{\rho+(1/6-\kappa)(3\rho-2+\delta)}.$$

Range II: $x^{1/3} < \ell \leq \frac{C}{\varepsilon}\sqrt{x} \log x$. Here $x + \ell^3 \leq 2\ell^3$, so (2.11) gives

$$P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^{\rho-1} \ell^{3\rho+\delta}.$$

Therefore,

$$\sum_{x^{1/3} < \ell \leq \frac{C}{\varepsilon}\sqrt{x} \log x} P_\ell(x) \ll_{\varepsilon, \rho, \delta} (\log x) x^{\rho-1} (\sqrt{x} \log x)^{1+3\rho+\delta} \ll_{\varepsilon, \rho, \delta} x^{-1/2+(5/2)\rho+\delta/2} (\log x)^{2+3\rho+\delta}.$$

Now choose $\rho, \delta > 0$ sufficiently small (depending on κ, β) so that

$$\rho + (1/6 - \kappa)(3\rho - 2 + \delta) < -\beta \quad \text{and} \quad -1/2 + (5/2)\rho + \delta/2 < -\beta.$$

This is possible since $\beta < \frac{1}{3} - 2\kappa$. Absorbing the logarithmic factors into $x^{o(1)}$ completes the proof of (2.10). $\square$

3. PROOF OF THEOREM 1.3

In this section, we give the proof of Theorem 1.3. We begin with some notations. For integers $m \geq 0$, we define

$$N_m(x) := \#\{p \leq x : f_\varepsilon(p) > m\} \quad \text{and} \quad a_m(x) := \frac{N_m(x)}{\pi(x)}.$$

Using the double counting trick, we have

$$\frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) = \sum_{m \geq 0} a_m(x). \quad (3.1)$$

3.1. Finite-dimensional equidistribution and limiting tail probabilities. The identity (3.1) shows that it suffices to understand the tail probabilities $a_m(x)$. For fixed m , the condition $f_\varepsilon(p) > m$ only depends on finitely many Legendre symbols $\chi_p(q)$ with $q \leq m$, so we begin by recording their equidistribution.

Lemma 3.1. *Let $q_1, \dots, q_r$ be distinct primes and let $\eta_1, \dots, \eta_r \in \{\pm 1\}$. Then*

$$\#\left\{p \leq x : p \notin \{q_1, \dots, q_r\}, \chi_p(q_i) = \eta_i \ (1 \leq i \leq r)\right\} = 2^{-r} \pi(x) + o(\pi(x)).$$

Proof. For primes $p \notin \{q_1, \dots, q_r\}$ we define

$$J(p) := \prod_{i=1}^r \frac{1 + \eta_i \cdot \chi_p(q_i)}{2}.$$

It is easy to see that $J(p) = 1$ if $\chi_p(q_i) = \eta_i$ for all i and $J(p) = 0$ otherwise. Expanding the product gives

$$J(p) = 2^{-r} \sum_{S \subseteq \{1, \dots, r\}} \left(\prod_{i \in S} \eta_i \right) \chi_p(q_S),$$

where $q_S = \prod_{i \in S} q_i$. For $S = \emptyset$ the summand is 1. For $S \neq \emptyset$, the map $p \mapsto \chi_p(q_S)$ is a nonprincipal quadratic Dirichlet character in the prime variable p . By the prime number theorem for Dirichlet characters [7, Corollary 5.29], we have

$$\sum_{p \leq x} \chi_p(q_S) = o(\pi(x)).$$

Summing $J(p)$ over primes $p \leq x$ and discarding the finitely many primes $p \in \{q_1, \dots, q_r\}$ gives

$$\sum_{p \leq x} J(p) = 2^{-r} \pi(x) + o(\pi(x)),$$

which is exactly the desired counting statement. $\square$

Fix $m \geq 1$, and list the primes $\leq m$ as

$$q_1 < q_2 < \dots < q_{\pi(m)}.$$

For a prime $p > m$ define

$$v_m(p) := (\chi_p(q_1), \chi_p(q_2), \dots, \chi_p(q_{\pi(m)})) \in \{\pm 1\}^{\pi(m)}.$$

Lemma 3.2. *Fix $m \geq 1$. Let*

$$A_m = \left\{ \sigma \in \{\pm 1\}^{\pi(m)} : v_m(p) = \sigma \text{ and } f_\varepsilon(p) > m \text{ for some prime } p > m \right\}.$$

Then the limit

$$\widehat{a}_m := \lim_{x \rightarrow \infty} a_m(x) \quad (3.2)$$

exists and equals $\frac{|A_m|}{2^{\pi(m)}}$.

Proof. We first show that for every prime $p > m$,

$$f_\varepsilon(p) > m \iff v_m(p) \in A_m.$$

In fact for primes $p, p' > m$, $v_m(p) = v_m(p')$ if and only if $\chi_p(q) = \chi_{p'}(q)$ for all primes $q \leq m$. This implies that $S_k(p) \geq \varepsilon k$ for all $k \leq m$ if and only if $S_k(p') \geq \varepsilon k$ for all $k \leq m$, i.e. $f_\varepsilon(p) > m$ if and only if $f_\varepsilon(p') > m$.

Now if $v_m(p) \in A_m$, then by definition of A_m , we see that there exists a prime $p' > m$ such that $v_m(p') = v_m(p)$ and $f_\varepsilon(p') > m$. By the discussion above, we have $f_\varepsilon(p) > m$. Conversely, if $f_\varepsilon(p) > m$, then $v_m(p) \in A_m$ by definition.

Therefore,

$$N_m(x) = \sum_{\sigma \in A_m} \#\{p \leq x : p > m, v_m(p) = \sigma\} + O(1),$$

where the $O(1)$ term accounts for the finitely many primes $p \leq m$. Applying Lemma 3.1 to each fixed $\sigma \in \{\pm 1\}^{\pi(m)}$ yields

$$\#\{p \leq x : p > m, v_m(p) = \sigma\} = \left(2^{-\pi(m)} + o(1)\right) \pi(x).$$

Summing over $\sigma \in A_m$ and dividing by $\pi(x)$ gives

$$a_m(x) = \frac{|A_m|}{2^{\pi(m)}} + o(1),$$

so the limit exists and equals $|A_m|/2^{\pi(m)}$. □

3.2. Uniform tail bounds and completion of the proof.

Proposition 3.3. *We have*

$$\lim_{M \rightarrow \infty} \limsup_{x \rightarrow \infty} \sum_{m > M} a_m(x) = 0.$$

Proof. Fix $\kappa \in (0, 1/6)$ and choose $\delta \in (0, 1)$ satisfying the condition given in the proof of Theorem 2.6. For any fixed M , take x sufficiently large so that $M < x^{1/6-\kappa}$ and thus

$$\sum_{m > M} a_m(x) = \sum_{M < m \leq x^{1/6-\kappa}} a_m(x) + \sum_{m > x^{1/6-\kappa}} a_m(x).$$

Note that if $f_\varepsilon(p) > m$, then $S_m(p) \geq \varepsilon m$. So we have a trivial inequality $a_m(x) \leq P_m(x)$. By (2.4), the first sum is bounded by

$$\sum_{M < m \leq x^{1/6-\kappa}} a_m(x) \ll_{\varepsilon, \kappa, \delta} \sum_{m > M} m^{-3+\delta}.$$

The second sum is $o(1)$ by Theorem 2.6. Hence

$$\limsup_{x \rightarrow \infty} \sum_{m > M} a_m(x) \ll_{\varepsilon, \kappa, \delta} \sum_{m > M} m^{-3+\delta}.$$

Letting $M \rightarrow \infty$ gives the claim. □

We are now ready to present the following.

Proof of Theorem 1.3. We first show that $\sum_{m \geq 0} \hat{a}_m < \infty$. In fact, we fix suitable constants $\kappa \in (0, 1/6)$ and $\delta \in (0, 1)$. For each fixed $m > 0$, taking x sufficiently large so that $m \leq x^{1/6-\kappa}$ and using (2.4) with the trivial inequality $a_m(x) \leq P_m(x)$, we obtain

$$\hat{a}_m = \lim_{x \rightarrow \infty} a_m(x) \leq \sup_{x \geq m^{6/(1-6\kappa)}} a_m(x) \ll_{\varepsilon, \kappa, \delta} m^{-3+\delta}.$$

Then the convergence of the series $\sum_{m \geq 0} \hat{a}_m$ follows. We write $c_\varepsilon := \sum_{m \geq 0} \hat{a}_m$.

Fix $M \geq 0$. Splitting (3.1) gives

$$\frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) = \sum_{m=0}^M a_m(x) + \sum_{m > M} a_m(x).$$

Taking $\liminf_{x \rightarrow \infty}$ and using nonnegativity yields

$$\liminf_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) \geq \sum_{m=0}^M \hat{a}_m.$$

Letting $M \rightarrow \infty$ gives

$$\liminf_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) \geq \sum_{m \geq 0} \hat{a}_m = c_\varepsilon.$$

For the $\limsup$, Proposition 3.3 implies that for every $\eta > 0$ there exists M such that $\limsup_{x \rightarrow \infty} \sum_{m > M} a_m(x) \leq \eta$. Hence

$$\limsup_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) \leq \lim_{x \rightarrow \infty} \sum_{m=0}^M a_m(x) + \eta = \sum_{m=0}^M \hat{a}_m + \eta \leq c_\varepsilon + \eta.$$

Since $\eta > 0$ is arbitrary, we get $\limsup \leq c_\varepsilon$. Combining with the $\liminf$ gives

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} f_\varepsilon(p) = c_\varepsilon.$$

This implies that $\sum_{p \leq x} f_\varepsilon(p) \sim c_\varepsilon \frac{x}{\log x}$ by the prime number theorem. $\square$

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